

LIVE CATTLE FUTURES

Spread & Butterfly Trading Under Fundamentals

Front of Curve vs Back of Curve — A Practical Framework

This document presents a professional framework for trading Live Cattle calendar spreads and butterflies using fundamental analysis. It distinguishes clearly between front-of-curve instruments (higher liquidity, tighter fundamental anchoring) and back-of-curve instruments (structural positioning, lower liquidity, longer time horizon).

Futures First — Internal Education Material
For Analyst Group Use

1. Live Cattle Curve Structure

Live Cattle (LE) trades on CME in six contract months per year: February (G), April (J), June (M), August (Q), October (V), and December (Z). This gives roughly 8 actively quoted contracts at any time, creating a curve spanning approximately 16 months forward. Tradeable spreads and flies typically run to the 7th or 8th contract before liquidity deteriorates materially.

Contract	Month	Zone	Liquidity	Fundamental Driver
C1 (front)	e.g. M26 (Jun)	FRONT	Highest	Cash basis, packer demand, immediate supply
C2	Q26 (Aug)	FRONT	High	Near-term feedlot placements, seasonal demand
C3	V26 (Oct)	FRONT	Good	Fall slaughter run, Q4 beef demand
C4	Z26 (Dec)	MID	Moderate	Holiday demand, year-end placement cycle
C5	G27 (Feb)	MID	Moderate	Winter slowdown, early placement data
C6	J27 (Apr)	BACK	Lower	Spring flush, early grilling season signal
C7-C8	M27+ (Jun+)	BACK	Thin	Long cycle placements, structural carry, COF outlook

The division front/back is not arbitrary — it reflects the fundamental information horizon. Front contracts price events that are knowable within weeks (cash kills, immediate feedlot capacity). Back contracts price structural expectations about placements made 5–9 months ago, which take time to resolve.

2. Fundamental Data Sources

Live Cattle is one of the most data-rich agricultural markets. Understanding which data source is relevant to which part of the curve is the core skill in fundamental spread trading.

2.1 USDA Cattle on Feed (COF)

Released monthly (third or fourth Friday). The single most important report for Live Cattle. Three key numbers:

- **Placements** — cattle placed on feed during the month. Drives back-of-curve contracts (cattle take 5–7 months to finish). Heavy placements now = heavy supply in C4–C7 range.
- **Marketings** — cattle sold to packers during the month. Drives front and mid-curve. Heavy marketings tighten current supply and often support nearby spreads.
- **On Feed Inventory** — total cattle in feedlots. High inventory = bearish pressure on curve. Low = bullish.

The key spread trade around COF: if placements are heavy in current month, consider selling back-of-curve spreads (supply coming). If marketings are heavy and placements light, buying front spreads is supported.

2.2 USDA Weekly Slaughter Data

Released every Friday afternoon for the prior week. Includes federally inspected (FI) slaughter numbers and average carcass weights. This is the highest-frequency fundamental input for front-of-curve spreads.

- FI slaughter significantly above or below expectations moves C1–C2 spreads within days.
- Carcass weights signal whether cattle coming to market are heavy (delayed marketings, potential short-term supply glut) or light (tight supply, supportive of nearby).

2.3 Cash Cattle Market

The negotiated cash market (fed cattle) in key regions (Texas/Oklahoma, Kansas, Nebraska, Colorado) is the fundamental anchor for front-month futures. Live Cattle futures converge to cash at expiration.

- Cash trading typically occurs Tuesday–Thursday each week.
- Packer bids vs asking prices signal packer profit margins and urgency of buying.
- **Basis** (cash minus nearby futures) is the cleanest short-term spread signal. Strengthening basis = buy nearby, sell deferred. Weakening = the opposite.

2.4 Boxed Beef Cutout Values

Daily USDA report on wholesale beef prices (Choice and Select cutouts). Packers buy live cattle and sell boxed beef — their margin is the spread between the two. High cutouts with firm cash = packer competition increases, supportive of front spreads.

- Choice-Select spread signals demand quality composition — wide = Choice premium demand, relevant for seasonal patterns around grilling season and holidays.

2.5 Cold Storage Reports

Monthly USDA report. High beef in cold storage = demand has lagged supply, bearish for front curve. Low storage = market is hand-to-mouth, bullish nearby.

2.6 Commitments of Traders (COT)

Weekly CFTC data on positioning by category: managed money, commercials (packers/feedlots hedging), swap dealers, small speculators. Most useful as a **crowding indicator** rather than a directional signal. Extreme managed

money net long → fade risk. Extreme commercial short → hedging pressure.

COT is more useful for back-of-curve timing than front — commercials hedge their anticipated production in deferred contracts, making positioning data a leading indicator for C4–C8 spread direction.

3. Front of Curve — C1 through C3 Spreads

Front spreads (C1–C2, C2–C3, and the C1–C2–C3 butterfly) are the most liquid, fastest-moving, and most directly fundamental instruments on the curve. They respond to data within days, sometimes hours.

3.1 What Drives Front Spreads

The hierarchy of inputs, in order of impact:

Priority	Data Source	Frequency	Impact on Spread	Direction Logic
1	Cash cattle market	Daily (Tue–Thu)	Immediate	Strong cash → C1 rallies vs C2 (basis tightens)
2	Weekly FI slaughter	Weekly (Fri)	1–3 days	High kills + heavy cwt = near-term supply glut, sell C1-C2
3	Boxed beef cutout	Daily	1–5 days	Rising Choice cutout → packer competition → buy nearby
4	COF Marketings	Monthly	3–10 days	Light marketings = tight near-term supply, buy C1 spread
5	Seasonal calendar	Structural	Weeks	Grilling season (Apr–Jul) demand premium narrows Q vs V

3.2 Seasonal Pattern: Front Spreads

Live Cattle has well-documented seasonal tendencies that represent the fundamental demand and supply cycle of beef production. For front spreads:

- **Q1 (Jan–Mar):** Post-holiday demand slowdown. Boxed beef typically weak. Front spreads often under pressure. C1–C2 spread tends to weaken as winter cattle hit the market.
- **Q2 (Apr–Jun):** Grilling season demand builds. Choice cutout typically rallies. Cash basis firms. Front spreads historically rally — the strongest seasonal period for M–Q spread.
- **Q3 (Jul–Aug):** Summer peak demand, heat stress slows cattle gains (lighter weights, tighter supply). C1 spreads often firm in early July, fade late July as heat-stressed cattle flood market.
- **Q4 (Oct–Dec):** Fall run — large numbers of cattle hit the market. V–Z spread often widens (deferred weakness). Holiday beef demand provides a partial offset for the front.

3.3 The M–Q–V Butterfly (Front Fly)

The M26–Q26–V26 butterfly is the most actively traded fly on the curve. It captures the curvature of supply/demand between the three summer/fall contracts. Fundamental interpretation:

- **Fly widens (positive)** when Q26 is cheap relative to its neighbours — i.e. supply is forecast to peak in August. This typically reflects heavy August-delivery placements 6 months prior.
- **Fly narrows or goes negative** when Q26 is expensive — August supply is seen tight, packers competing aggressively for Q-month cattle.

Key: the front fly is a supply timing instrument. Heavy COF placements in Oct–Nov (6 months before Q) forecast a wide fly. Light placements = narrow or negative fly.

3.4 Risk Profile — Front of Curve

- High liquidity = tight bid-ask, executable size. Multiple market participants anchor price to fundamentals quickly.

- Data risk is high — a single weekly slaughter number or cash trade can move C1–C2 spread 15–30 cents in one session.
- Carry risk is low — holding for days to weeks is feasible.

CAUTION: Front spreads are crowded by commercials. Packers and feedlots hedge here continuously. Do not fight packer hedging pressure without a strong fundamental catalyst.

4. Back of Curve — C4 through C8 Spreads

Back-of-curve spreads (Z26–G27 and beyond, flies centred on G27, J27, M27) operate on a fundamentally different time horizon. The cattle being priced in these contracts are **currently in feedlots or not yet placed**. Information is softer, liquidity is thinner, and the holding period is longer.

4.1 What Drives Back Spreads

Priority	Data Source	Frequency	Lead Time to Effect	Direction Logic
1	COF Placements (monthly)	Monthly	5–7 months	Heavy placements now = supply in C5–C7; sell those spreads
2	Feeder cattle prices & availability	Weekly	4–8 months	High feeder prices = fewer placements = tight back-curve supply
3	Feed costs (corn, DDGs)	Daily	4–8 months	High feed costs = feedlots delay/reduce placements = bullish back
4	COT (commercial positioning)	Weekly	Weeks–months	Heavy commercial short in deferred = supply expected; fade extremes
5	Drought index / pasture conditions	Weekly	6–18 months	Drought = forced liquidation of breeding herd = tighter 2-yr supply
6	Cattle cycle position	Structural	Years	Expansion vs contraction phase determines structural carry shape

4.2 The Cattle Production Cycle

Live Cattle has a biological production lag that is unique among commodity markets. Understanding this lag is essential for back-of-curve trading:

- **Placement to slaughter:** 5–7 months for fed cattle. This is the most important lag. A record heavy COF placement month creates predictable supply pressure 5–7 months forward.
- **Breeding decision to feeder availability:** 9–12 months. Changes in cow-calf operator behaviour (driven by drought, beef prices, input costs) take 9–12 months to show in feeder supply.
- **Herd rebuilding vs liquidation:** Multi-year cycle. When cattle inventory is at lows and prices are high, producers retain heifers to rebuild — temporarily reducing slaughter supply but guaranteeing more supply 2–3 years forward.

Back-of-curve spread position = bet on whether the market has correctly priced the known biological lag. This is a genuine informational edge if you track placements, feeder prices and feed costs rigorously.

4.3 Carry Structure in Back of Curve

Unlike energy or metals, Live Cattle does not have a storage-based carry model. However, a structural cost-of-carry concept exists:

- Feedlot cost of gain — the daily cost of feeding cattle forward. This creates a theoretical minimum carry between contracts (approximately \$0.50–1.50/cwt per month depending on feed costs).
- When back spreads trade **below this theoretical carry**, the market is pricing future supply abundance — sell back spreads.
- When back spreads trade **above carry**, expected tightness is priced — buy back spreads or wait for mean-reversion.

4.4 Risk Profile — Back of Curve

- Thin liquidity = wider bid-ask, limited size. Positions must be sized smaller and held for weeks to months.
- Fundamental edge is larger because fewer participants analyse the information correctly or have the patience to hold.
- Data risk is lower per event — no single report moves back spreads the way weekly slaughter moves C1–C2. But accumulation over months matters.

CAUTION: Thin liquidity creates execution risk. A 5-lot position in C7–C8 spread may be manageable on entry but difficult to exit quickly if the fundamental view is wrong.

5. Front vs Back — Key Differences

Dimension	Front of Curve (C1–C3)	Back of Curve (C4–C8)
Primary driver	Cash market, weekly kills, boxed beef	COF Placements, feeder prices, feed costs
Data frequency	Daily to weekly	Monthly (COF) + structural
Info-to-price lag	1–5 trading days	Weeks to months
Holding period	Days to 3 weeks	3 weeks to 3 months
Liquidity	High — executable 20–100+ lots	Low–moderate — 5–20 lots typical
Dominant participants	Packers, funds, spread desks	Feedlot hedgers, commercial shorts
Main seasonal bias	Apr–Jun bullish (grilling demand)	Depends on placement cycle 6 months prior
Fly character	Supply timing — reacts to placement data	Structural carry curvature — longer-term view
Edge source	Speed of data interpretation + basis tracking	Patience + production cycle modelling
Main risk	Data surprise, packer behaviour, crowding	Illiquidity, slow convergence, fundamental error

6. Practical Trade Construction

6.1 Front Spread Trade — Step by Step

Example: Positioning in M26–Q26 spread ahead of COF report

- **Step 1 — Hypothesis:** October COF placements were light (published in November). This suggests tight supply for Q26 (August delivery). Expect M–Q spread to narrow (Q gets bid relative to M).
- **Step 2 — Confirmation:** Check cash basis trend. Is cash strengthening in nearby region? Are carcass weights declining (supply tightening sign)?
- **Step 3 — Entry:** Enter 2–3 weeks after COF if market has not fully priced the tightness. Use limit orders — front spreads have enough liquidity to be patient.
- **Step 4 — Exit thesis:** Spread reverts to fair value, OR next COF shows heavy placements reversing the thesis, OR cash basis weakens materially.
- **Step 5 — Risk:** Define max loss (e.g. 40–50 cents adverse move) before entry. Front spreads can gap on weekly data releases.

6.2 Back Spread Trade — Step by Step

Example: Positioning in G27–J27 spread based on feeder price signal

- **Step 1 — Hypothesis:** Feeder cattle prices have rallied 15% in past 3 months. Feedlot operators are placing fewer head. G27–J27 spread should widen as supply tightens for Feb–Apr 2027.
- **Step 2 — Confirmation:** Cross-reference with COF placements trend (are they declining?). Check drought monitor — is pasture deterioration forcing feeder retention?
- **Step 3 — Entry:** Enter over several days, scaling in. Liquidity is thin — use wide limit orders, accept partial fills.
- **Step 4 — Exit thesis:** Feeder prices stabilise/decline (placements resume), OR COF placements reverse, OR position approaches target after 6–10 weeks.
- **Step 5 — Risk:** Size 30–40% of what you would use on front spreads. Illiquidity means exit cost is real.

6.3 Combining Fundamentals with Order Flow

The most powerful approach combines fundamental direction with order flow confirmation. Specifically for Live Cattle spreads:

- When COF placements are fundamentally bullish for a spread AND large buy iceberg orders appear in that spread — the institutional execution confirms the thesis is known to professional capital.
- When fundamentals are bullish but only sell icebergs appear — re-examine the thesis. Large sell flow in a "bullish" spread may indicate hedging by producers who know more.
- The absence of iceberg activity in back spreads is normal — do not wait for flow confirmation that may never come.

Rule of thumb: fundamentals give you the direction, order flow tells you whether the market is ready to move.

7. Common Mistakes in Fundamental Spread Trading

Treating the spread like an outright

A bullish fundamental outlook on Live Cattle does not automatically mean buy front spreads. Ask: which contract is specifically under- or over-supplied relative to its neighbour? Outrights move on macro sentiment. Spreads move on relative supply/demand.

Ignoring the data release calendar

COF is released monthly. Weekly slaughter every Friday. Holding a front spread through an unexpected COF or slaughter number without a stop is high-risk. Know the calendar and size accordingly.

Overtrading back-of-curve on front data

Weekly slaughter data is largely irrelevant to G27 or J27 spreads. Using short-term data to justify long-term positions leads to churning and mis-sized risk.

Ignoring packer behaviour

Packers hedge forward production continuously. In deferred contracts, commercial short positioning is structurally present — it does not always mean sell. Understand context before fading commercial shorts.

Confusing carry with edge

When a back spread is trading at theoretical carry, there is no edge either way. Edge comes from deviation from carry driven by known fundamental information — heavy/light placements, drought, feeder scarcity.

Expecting fast convergence in back spreads

A correct fundamental thesis on G27–J27 may take 8–12 weeks to materialise. Exiting early because the position is flat for 3 weeks is a common error. Define exit by thesis change, not impatience.

8. Key Data Release Schedule

Release	Source	Frequency	Time (CT)	Primary Impact
Cattle on Feed	USDA NASS	Monthly (3rd/4th Fri)	2:00 PM	All spreads, especially C4–C7
Weekly Slaughter	USDA AMS	Weekly (Fri ~4 PM)	~4:00 PM	C1–C3 spreads, cash basis
Boxed Beef Cutout	USDA AMS	Daily (Mon–Fri)	11 AM / 3 PM	Front spreads, packer margin
Cash Cattle Report	USDA AMS	Daily (Tue–Thu)	Ongoing	C1 basis, front-month direction
Cold Storage	USDA NASS	Monthly (last Fri)	2:00 PM	Mid-curve supply assessment
Drought Monitor	NDMC/USDA	Weekly (Thu)	Ongoing	Long-cycle supply, feeder availability
Commitments of Traders	CFTC	Weekly (Fri ~3:30 PM)	~3:30 PM	Positioning crowding, back-curve signal

The fundamental approach to Live Cattle spread trading requires patience, discipline in data tracking, and a clear separation between front-curve tactical positions and back-curve structural positions. Neither approach is superior —

they serve different risk/reward profiles and different trader temperaments. The front curve rewards speed and data interpretation. The back curve rewards conviction and knowledge of the production cycle.

The intersection of both — when fundamental thesis aligns with observable institutional order flow — represents the highest-probability setup available in this market.